**Proof.** A simple combinatorial argument implies that the graph of every 2-diagram is 3-connected (it is simple and planar by construction). Thus the theorem follows from Steinitz' Theorem 4.1, together with the construction of Schlegel diagrams of 3-polytopes. $\square$

**Proposition 5.9.** *If a diagram $\mathcal{D}$ is Schlegel, then $\mathcal{D}$ is a regular subdivision of $F = |\mathcal{D}|$. The converse is true in the case where $F$ is a simplex, but not in general.*

We omit the proof — see Exercises 5.2(ii) and 5.7.

There are other properties that separate Schlegel diagrams from some non-Schlegel diagrams. For example, let $\mathcal{D}$ be a $d$-diagram, and let $L$ be the "reconstructed" face lattice of the $(d+1)$-polytope $P$, if it exists. The combinatorial information about $\mathcal{D}$ is then contained in the pair $(L, F)$, where $F$ is a distinguished coatom of $L$. Now we call a diagram *invertible* if for every coatom $F'$ of $L$, there is a $d$-diagram corresponding to the pair $(L, F')$. We know that every Schlegel diagram is invertible. It turns out that a part of the non-Schlegel diagrams is not invertible.

Similarly, we say that $\mathcal{D}$ has a *combinatorial polar* if there is a diagram whose combinatorial data are given by the pair $(L^{op}, A)$, where $A$ is a vertex of $\mathcal{D}$ (corresponding to an atom of $L$, and thus to a coatom of $L^{op}$). Again, every Schlegel diagram has a polar diagram (which is Schlegel), and some non-Schlegel diagrams have polars, some do not.

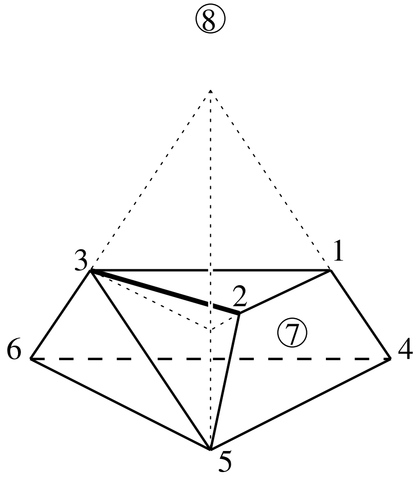
## 5.4 Three Examples

Examples of interesting Schlegel diagrams, and of 3-diagrams that are not Schlegel diagrams, are not too hard to come by. In the following we will describe three strikingly simple examples, where two are due to Barnette [47], and one is from Schulz [487]. See also Ewald [201, Sect. IV.4] for the "classical" examples of Brückner and Barnette (see the notes), and Schulz [488] for some other interesting constructions.

From general theory (namely the technique of "Gale diagrams," which we will soon get to) one can see that every $d$-diagram with at most $d + 4$ vertices is always Schlegel. Therefore, the minimal counterexamples that we can hope for are 3-diagrams with 8 vertices.

Here we go: the following construction produces *a 3-diagram that is not combinatorially equivalent to the Schlegel diagram of any 4-polytope.*

**Example 5.10 (Schulz' 3-diagram).** [487]
We start with a 3-polytope $Q$ with 6 vertices, labeled $1, 2, \ldots, 6$ in our drawing. The 3-polytope can be realized by starting with a tetrahedron, cutting off the top to get a triangular prism, and then cutting off an extra triangle $[2, 3, 5]$ as in the figure, where $3, 5$ were vertices of the truncated prism, and 2 was on an edge of it.




Now we choose extra points: 7 in general position inside $Q$, but outside the tetrahedron $[2,3,5,6]$, and 8 above the top of the original pyramid, so that all facets of $Q$, except for the base $[4,5,6]$, can be "seen" from 8.

The diagram $\mathcal{D}_1$, based on the tetrahedron $G = [4,5,6,8]$, consists of the following ten 3-polytopes in $\mathbb{R}^3$, and their faces:

$A$: $[2,3,5,6,7]$, a bipyramid over the triangle 357,
$B, C$: $[1,2,3,7]$ and $[4,5,6,7]$, two tetrahedra, and
$D, E$: $[1,2,4,5,7]$ and $[1,3,4,6,7]$, two square pyramids,

where $B, C, D, E$ together cover the interior of $Q \backslash A$, by using 7 as a cone point,

$F, K, L$: $[1,2,3,8]$, $[2,3,5,8]$ and $[3,5,6,8]$, three tetrahedra, and
$H, I$: $[1,2,4,5,8]$ and $[1,3,4,6,8]$, two square pyramids,

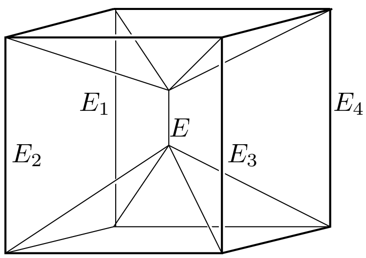
where $F, K, L, H, I$ together cover the interior of $G \backslash Q$, by using 8 as a cone point.

It is easy to see now that this is a valid 3-diagram. Why is it not combinatorially equivalent to a Schlegel diagram? Assuming it is, then there is a 4-polytope $P$ whose vertex set we can label $1, 2 \ldots, 8$, and whose Schlegel diagram is equivalent to $\mathcal{D}_1$.

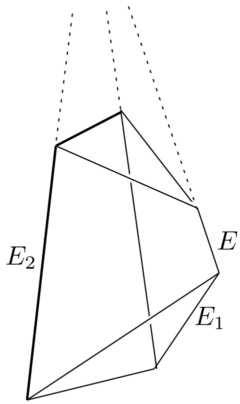
Now all vertices of $Q$ are contained in the union of the two 2-faces $[1,2,4,5]$ and $[1,3,4,6]$, which share an edge $[1,4]$. Thus the affine span $R := \mathrm{aff}\{1,2,3,4,5,6\}$ of the vertices of $Q$ has dimension 3 in $\mathbb{R}^4$. Furthermore, the triangles $[2,3,5]$ and $[3,5,6]$ are contained in $R$. Thus the facet $A$ has two 2-faces in $R$, and therefore it must be contained in $R$. Thus the vertex 7 and all of its neighbors are contained in $R \subset \mathbb{R}^4$, which is a contradiction (to the fact that different facets of $P$ must span different hyperplanes in $\mathbb{R}^4$, or to Lemma 3.6). $\square$

Our next example, $\mathcal{D}_2$, is a Schlegel diagram for which *the total space* $F = |\mathcal{D}_2|$ *cannot be prescribed.* This shows that for a 4-polytope, we *cannot prescribe the shape of a facet* — in contrast to the situation for 3-polytopes, as discussed in Section 4.4.

**Example 5.11 (Barnette's first diagram).** [47]
For this, we consider the Schlegel diagram for the prism over the square pyramid $I \times Pyr_3$. This Schlegel diagram can be constructed from a regular cube, and two vertices on the vertical axis of symmetry.




Now consider any 3-diagram that is combinatorially equivalent to this. Then the lines determined by the edges $E_1$, $E_2$, and $E$ belong to a *pencil of lines*, that is, either they are all parallel, or they have a common point of intersection. To see this, just consider the plane $R = \text{aff}(E_1 \cup E_2)$ determined by the lines $E_1$ and $E_2$, and the intersection point with the line $\text{aff}(E)$, which may be "at infinity." Using that $\text{aff}(E \cup E_1)$ is a plane, we get that the intersection point is contained in the line $\text{aff}(E \cup E_1) \cap \text{aff}(E_1 \cup E_2) = \text{aff}(E_1)$, and similarly it is contained in the line $\text{aff}(E_2)$.




From symmetric arguments for $E_i$, $E_{i+1}$, and $E$, we see that for every 3-diagram that is combinatorially equivalent to the given one, the four lines $E_1, E_2, E_3, E_4$ are parallel, or else they intersect in a common point.

Thus if we start with a "skew" combinatorial cube that does not satisfy this (such a cube is easy to get), then we cannot even complete it to a

3-diagram that is isomorphic to the given one. In particular, such a skew cube is never a facet of a prism over a square pyramid. □

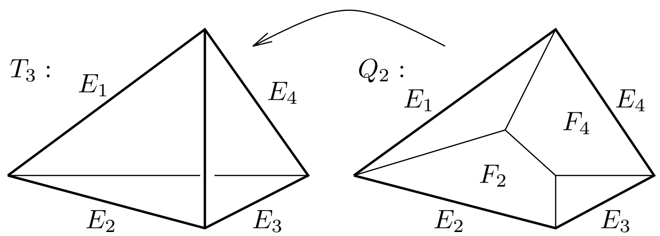
Steinitz' theorem can be stated in the following way: any "reasonable" cell decomposition of the 2-sphere (technically, we can consider regular CW-complexes with the intersection property; see [96, Sect. 4.7]) can be realized as the boundary complex of a convex 3-polytope. It is a weaker statement that, after deleting the interior of a 2-face, the rest can be realized by a 2-diagram.

This formulation of Steinitz' theorem has an obvious generalization to 3-spheres and 4-polytopes — and Schulz' Example 5.10 showed that this generalization is false. In fact, it is not hard to see that any $d$-diagram defines a "reasonable" cell decomposition of the $d$-sphere. Thus, diagram $\mathcal{D}_1$ represents a 3-sphere that can be realized by a 3-diagram, but not by a 4-polytope.

Our next example shows that some 3-spheres cannot even be represented by a 3-diagram (at least not with a specified simplex as its base).

**Example 5.12 (Barnette's topological diagram).** [47]
Barnette's example, $\mathcal{D}_3$, is a "curved," *topological* 3-*diagram that cannot be straightened at all.* This suggests that an effective combinatorial characterization of $d$-diagrams and Schlegel diagrams is too much to hope for.

For this, we start with a tetrahedron $T_3$. Into this tetrahedron, we glue a subdivision of a quadrangle $Q_2$, as indicated in our drawing, in such a way that the boundary $E_1, E_2, E_3, E_4$ of the quadrangle is identified with a circuit of four edges $E_1, E_2, E_3, E_4$ on the boundary of the tetrahedron, and the interior of $Q_2$ gets mapped into the interior of $T_3$, along a curved surface (which you may think of as a soap film bounded by the four edges $E_i$).




The curved quadrangle partitions the interior of $T_3$ into two "3-cell regions." Into each of them we place a new vertex and then perform a "stellar subdivision": each vertex is joined to all the faces on the boundary of the respective 3-cell, so each 3-cell is replaced by the (topological) pyramids over four triangles and two quadrangles each, and their faces.

But we find that this "topological 3-diagram" is not realizable. In fact, any two realizations of the tetrahedron are equivalent. Now we try to realize the quadrangles $F_2$ and $F_4$ in $Q_2$ by planar convex quadrangles. Then the plane $H_2 := \mathrm{aff}(F_2)$ contains $E_2$ and a (unique) point of $E_4$. Similarly, the plane $H_4 := \operatorname{aff}(F_4)$ then contains $E_4$ and a point of $E_2$. Thus the intersection $H_2 \cap H_4$ connects a point on $E_2$ with a point on $E_4$. This means that the four points of $F_2$ are not in convex position in $T_3$. $\square$

# Notes

Schlegel diagrams are the most direct, and probably the most effective, tool to visualize 4-dimensional objects. Of course, there is a certain amount of training necessary to develop the geometric intuition. Don't let yourself get discouraged, not even by statements like the following:

> Here, however, a word of warning may be in order: do *not* try to visualize $n$-dimensional objects for $n \geq 4$. Such an effort is not only doomed to failure—it may be dangerous to your mental health. (If you do succeed, then you are in trouble.) To speak of $n$-dimensional geometry with $n \geq 4$ simply means to speak of a certain part of algebra. (Chvátal [158, p. 252])

This is wrong, and even Chvátal acknowledges the fact that the correspondence between intuitive geometric terms and algebraic machinery can be used in both ways [158, p. 250].

The technique of Schlegel diagrams was already used extensively in work of Brückner [137] early this century, where the distinction between Schlegel diagrams and 3-diagrams was not made.

There are also *simplicial* examples known of 3-diagrams that are not Schlegel diagrams, and not even combinatorially equivalent to such. The first one was described by Grünbaum's abstract [251], which started the subject. The first non-Schlegel 3-diagram with 8 vertices was found by Grünbaum & Sreedharan [260], showing that one of Brückner's 3-diagrams does not, as Brückner thought, represent the combinatorial type of a 4-polytope. It is now known as the *Brückner sphere* [252, p. 222]. A second example of a simplicial 3-diagram that is not equivalent to a Schlegel diagram — the *Barnette sphere* — was found by Barnette [39] a little later. There are also simplicial 3-spheres that can be represented by topological diagrams (as in our Example 5.12), but not by straight 3-diagrams. Both kinds of examples are nicely presented in Ewald [201, Sect. IV.4 and IV.5].

However, there is something special happening in the case of *simple* diagrams. In fact, *every* simple $d$-diagram with $d \geq 3$ *is* the Schlegel diagram of a $(d+1)$-polytope. Thus, there are also things *true* in 3-space that are false in 2 dimensions. This was proved by Whiteley [563], and in an even stronger version by Rybnikov [470]. (They use a quite general setting for "liftability"; see Crapo & Whiteley [165, 166, 167].) Earlier results of Davis [180] and Aurenhammer [26] did not include the "boundary" conditions that pose extra constraints (as in Exercise 5.7).